IE08

Services, Infrastructure, External Sector, Employment and Skills

About 3-4 questions a paper out of the 8 Indian Economy questions, plus overlap with Indian Geography. Documented repeats: the Golden Quadrilateral and the corridor crossing at Jhansi (2013, 2016, 2021, 2023, 2024 - the most repeated infrastructure fact in the exam) and the Work Participation Rate (2021, 2024). Skill development architecture was asked in 2018 and 2021.

This is the widest chapter in Indian Economy - services, roads, railways, ports, airports, trade, jobs. Do not try to understand it as economics. Treat it as four card decks: programme-to-ministry, programme-to-launch-year, account-to-item, and survey-to-number. That is the shape in which RPSC asks all of it.

Services, Infrastructure, External Sector, Jobs

Services

- Largest share of GVA, roughly 55 per cent
- India among the largest exporters of commercial services
- Services exports FY 2024-25: US$ 383.51 billion

Power

- Electricity Act 2003 - unbundling
- UDAY, November 2015 - DISCOM finances
- RDSS, 2021
- Panchamrit at COP26 - 500 GW non-fossil by 2030, net zero 2070
- PM Surya Ghar, 13 February 2024 - 300 free units

Roads and rail

- Golden Quadrilateral 5,846 km
- NS and EW corridors cross at Jhansi
- Bharatmala Phase-I, 2017 - about 34,800 km
- Eastern DFC Ludhiana-Sonnagar, Western DFC JNPT-Dadri, DFCCIL 2006
- Vande Bharat 15 February 2019; Kavach

Ports, air, telecom

- Sagarmala 2015 - Ports, Shipping and Waterways
- Major Port Authorities Act 2021 replaced the 1963 Act
- UDAN 21 October 2016, first flight Shimla-Delhi April 2017
- Telecommunications Act 2023 replaced the 1885 Act
- BharatNet; 5G launched 1 October 2022

Planning and PPP

- PM Gati Shakti 13 October 2021, nodal DPIIT
- Seven engines; National Logistics Policy 17 September 2022
- NIP - Atanu Chakraborty task force, Rs 111 lakh crore
- BOT Toll, BOT Annuity, HAM (40 per cent), TOT
- VGF by DEA; Kelkar Committee 2015

External sector

- BoP = current account + capital account
- FDI 10 per cent threshold; FPI is hot money
- Reserves = FCA + gold + SDR + reserve tranche
- LERMS 1992, unified March 1993, Article VIII August 1994
- WTO 1 January 1995, Geneva, 166 members

Employment and skills

- About 90 per cent of workers are unorganised
- e-Shram, 26 August 2021, Labour Ministry
- PLFS from April 2017; WPR = employed / population
- MSDE created 9 November 2014; NSDC 2008, 49:51
- NSQF December 2013, 10 levels; PMKVY 15 July 2015
- Gig worker first defined in the Code on Social Security

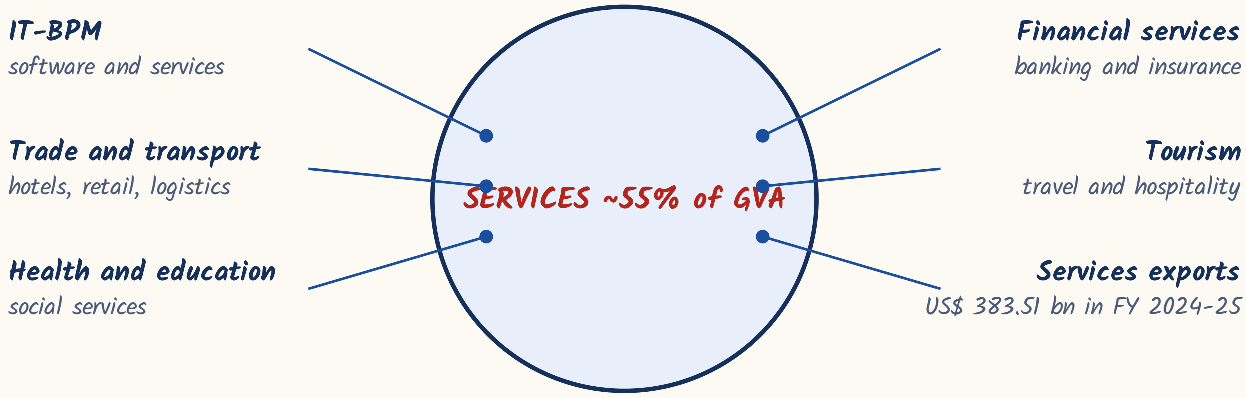
2020

Services – the sector that carries the economy

Services are the largest part of India's economy by output but not by employment. That gap is the whole point of the sector and it is where the exam question usually sits. Services contribute the largest share of India's gross value added, roughly 55 per cent, while agriculture still employs far more people than it produces. Within services, the export story is what makes India distinctive – India is among the world's largest exporters of commercial services, and software and business services do most of that work.

- Services contribute the largest share of India's gross value added – roughly 55 per cent.
- India's services exports in FY 2024-25 were US$ 383.51 billion.
- The big services sub-sectors: IT-BPM, financial services, trade and transport, tourism, healthcare and education.
- Services exports need no shipping and no port – which is why India's services surplus offsets its merchandise deficit.
- The paradox to remember: highest share of output, not the highest share of employment.

Inside the services sector - largest in output, not in jobs



TRAP

Share of gross value added is not the same as share of employment. If an option says services employ the largest number of Indians, it is wrong - agriculture still does. Output share and employment share are two different questions.

Power - reforms, UDAY and the renewable targets

India's power problem was never generation. It was distribution - state DISCOMs selling below cost and running up losses. So every power reform since 2003 has been aimed at the distribution company. Learn the reform ladder in order: unbundling first, then a bailout, then a conditional scheme.

- Electricity Act, 2003 unbundled generation, transmission and distribution into separate entities.
- UDAY - Ujwal DISCOM Assurance Yojana - was launched in November 2015 to restore DISCOM finances, with states taking over DISCOM debt.
- The Revamped Distribution Sector Scheme (RDSS) followed in 2021, tying central funds to loss-reduction milestones.
- UDAY is about power DISTRIBUTION companies, not generation companies. That is the trick in the question.

Renewables and the Panchamrit commitments

- At COP26 in Glasgow in 2021, India announced the 'Panchamrit' - five commitments.
- The two that get asked: 500 GW of non-fossil energy capacity by 2030, and net zero by 2070.
- PM Surya Ghar Muft Bijli Yojana was launched on 13 February 2024.
- It targets rooftop solar for one crore households with up to 300 units of free electricity a month.
- Remember the number pair - 500 GW and 2070 for Panchamrit; one crore households and 300 units for PM Surya Ghar.

Roads - the most repeated infrastructure topic in the exam

If you learn only one thing from this chapter, learn the highway corridors. The Golden Quadrilateral joins the four metros. Two long corridors cross the country north-south and east-west, and they intersect at Jhansi. RPSC has asked this cluster in 2013, 2016, 2021, 2023 and 2024. It is the single most repeated geography-economy fact in the paper.

The National Highways Development Project corridors

Corridor	Route	Length
Golden Quadrilateral	Delhi - Mumbai - Chennai - Kolkata - Delhi	5,846 km
North-South corridor	Srinagar to Kanyakumari	About 4,076 km
East-West corridor	Silchar to Porbandar	About 3,640 km
Crossing point	North-South meets East-West	Jhansi, Uttar Pradesh

- Bharatmala Pariyojana Phase-I was approved in 2017 for about 34,800 km.
- It covers economic corridors, inter-corridor routes, border and international connectivity roads, coastal roads and expressways.
- Bharatmala is implemented mainly through the National Highways Authority of India.
- Bharatmala came after the NHDP - do not treat them as the same programme.

ASKED IN RAS

Asked in RAS Pre 2021 and again in 2024 - the total length of the Golden Quadrilateral is about 5,846 km. Learn the exact figure, because the wrong options are all near it.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2023 - the North-South and East-West corridors of the NHDP intersect at Jhansi. One word, one mark, asked twice.

TRAP

Srinagar-Kanyakumari is North-South. Silchar-Porbandar is East-West. Students reverse them under pressure. Silchar is in Assam, in the east - so Silchar starts the East-West corridor.

Railways, ports, airports and telecom

Four transport and communication systems, four sets of programme facts. What RPSC asks is always the same three things - which ministry, which year, and which older law was replaced. Learn them in that shape.

- Dedicated Freight Corridors - Eastern DFC runs Ludhiana to Sonnagar, Western DFC runs Jawaharlal Nehru Port to Dadri.
- DFCs are implemented by DFCCIL, incorporated in 2006.
- The Western DFC is the spine along which the Delhi-Mumbai Industrial Corridor is planned.
- The first Vande Bharat Express, originally Train 18, was flagged off on 15 February 2019 on the New Delhi-Varanasi route.
- Kavach is the indigenous train collision avoidance system.

Ports, airports, telecom and digital infrastructure

- Sagarmala, launched in 2015, is the port-led development programme of the Ministry of Ports, Shipping and Waterways.
- The Major Port Authorities Act, 2021 replaced the Major Port Trusts Act, 1963 and gave major ports greater autonomy.
- UDAN - Ude Desh ka Aam Naagrik - the Regional Connectivity Scheme, was launched on 21 October 2016.
- The first UDAN flight was Shimla to Delhi in April 2017.
- Under UDAN, fares on RCS seats are capped and airlines receive viability gap funding.
- The Telecommunications Act, 2023 replaced the Indian Telegraph Act, 1885 and the Indian Wireless Telegraphy Act, 1933.
- TRAI was set up under the TRAI Act, 1997.
- BharatNet, originally the National Optical Fibre Network of 2011, aims to take broadband to every gram panchayat.
- 5G services in India were launched on 1 October 2022.

PM Gati Shakti and the National Infrastructure Pipeline

These two are often confused. The National Infrastructure Pipeline is a money list - a pipeline of projects worth a stated sum. PM Gati Shakti is a map - a GIS platform on which ministries plan their projects together so that a road is not dug up a month after it is laid. One is finance, one is coordination. Fix that difference and the statement questions become easy.

The infrastructure programme chronology - a favourite ordering question

Electricity Act, 2003 → Sagarmala and UDAY, 2015 → UDAN, October 2016 → Bharatmala Phase-I, 2017 → National Infrastructure Pipeline, 2019-20 → PM Gati Shakti, 13 October 2021 → National Logistics Policy, 17 September 2022

- PM Gati Shakti National Master Plan was launched on 13 October 2021.
- It is a GIS-based digital platform built with BiSAG-N, integrating 16 ministries.
- The nodal department is DPIIT - the same department that runs Make in India and the state business-reform ranking.
- Its institutional structure is a Network Planning Group under an Empowered Group of Secretaries.
- Budget 2022-23 named seven engines of Gati Shakti - roads, railways, airports, ports, mass transport, waterways and logistics infrastructure.
- Power transmission is NOT one of the seven engines. That is the standard wrong option.
- The National Logistics Policy was released on 17 September 2022.

- The National Infrastructure Pipeline task force was chaired by Atanu Chakraborty.
- It listed projects worth about Rs 111 lakh crore for FY2020 to FY2025.
- NIP projects are showcased for investors on the India Investment Grid.
- NIP is a pipeline of projects and their funding - it is not a physical corridor or a digital platform.

PPP models - BOT, HAM and TOT

A public-private partnership question is always really a question about who carries the risk. Build the road, then ask - who loses money if traffic is thin? In BOT (Toll) it is the private concessionaire, because he lives off the toll. In BOT (Annuity) it is the government, because it pays a fixed annuity whatever the traffic. HAM splits the difference. TOT is not construction at all - it is the sale of the earning right on a road already built. Understand the risk question and you never have to memorise the models.

The PPP models used in Indian highways

Model	How it works	Who bears traffic risk
BOT (Toll)	Private party builds, operates and collects toll, then transfers	The concessionaire
BOT (Annuity)	Private party builds and operates; the authority pays a fixed annuity	The authority (government)
Hybrid Annuity Model (HAM)	Government funds 40 per cent of project cost during construction; the rest comes back to the developer as annuities	Shared - the authority keeps toll risk
TOT - Toll Operate Transfer	An already operating highway is leased to a private operator for a lump sum; monetisation, not construction	The operator, for the lease period
EPC - Engineering, Procurement, Construction	Government pays the contractor to build; government owns and operates	Not a PPP model at all

BOT (Toll) versus Hybrid Annuity Model

BOT (Toll)	Hybrid Annuity Model
Developer raises the entire project cost	Government funds 40 per cent during construction
Developer recovers money from toll collection	Developer recovers the balance 60 per cent as annuities
Traffic risk fully with the developer	Traffic risk stays with the authority
Attractive only on busy corridors	Works on thin-traffic corridors too, which is why it revived road building

- Remember the number: HAM = 40 per cent government funding during construction.
- EPC is NOT a PPP model - the government pays and the government owns. This is the classic 'which is not' option.
- Viability Gap Funding is administered by the Department of Economic Affairs, Ministry of Finance.
- The Kelkar Committee of 2015 reviewed and recommended revamping the PPP model in India.

The external sector - balance of payments, FDI, FPI and reserves

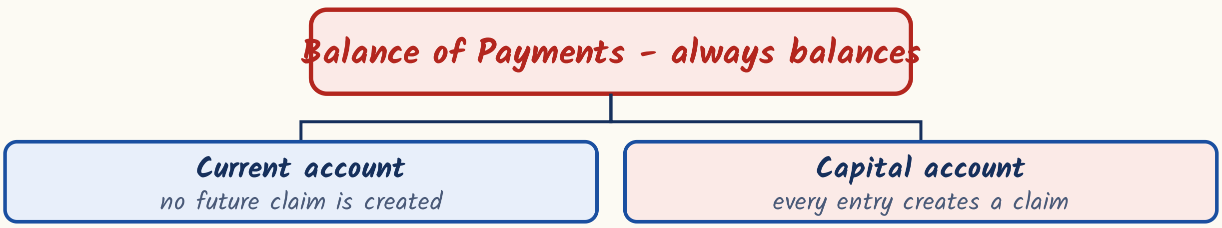
The balance of payments is simply a record of every transaction between residents of India and the rest of the world. It has two halves. The current account records things that are consumed now - goods, services, income and transfers. The capital account records claims and liabilities - who owns what. And here is the sentence RPSC keeps testing: the balance of payments always balances, by construction. A 'BoP deficit' means a current account deficit financed by drawing down reserves or by capital inflows.

Current account versus capital account

Current account	Capital account
Merchandise trade - exports and imports of goods	Foreign direct investment
Invisibles - services, primary income, transfers	Foreign portfolio investment
Remittances from Indians abroad	External commercial borrowings
No creation of a future claim or liability	Banking capital, NRI deposits, loans - each creates a claim
A deficit here means we consumed more than we produced	A surplus here finances the current account deficit

- External commercial borrowings are a capital account item, not a current account item. This exact option has been used as the odd one out.
- Invisibles = services + primary income + transfers. Software exports and remittances sit here.
- The BoP always balances; a deficit is financed by reserves or by capital inflows.
- India typically runs a merchandise trade deficit and a services and remittances surplus.

The two halves of the balance of payments



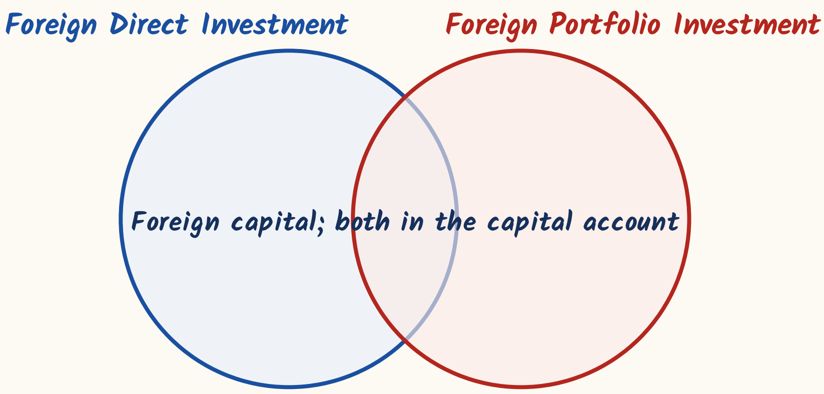
- Merchandise trade
- Services - invisibles
- Primary income
- Transfers and remittances

- FDI - lasting stake
- FPI - portfolio money
- ECBs - foreign loans
- Banking, NRI deposits

FDI versus FPI - learn this as a pair

Foreign Direct Investment	Foreign Portfolio Investment
Implies a lasting interest in the enterprise	A purely financial stake, no management interest
Conventionally 10 per cent or more of equity	Below 10 per cent of equity
Brings technology, management and market access	Brings only capital
Relatively stable and long-term	Volatile - the classic 'hot money'
Enters through automatic or approval routes under FEMA	Enters through registered foreign portfolio investors under SEBI rules

FDI and FPI



The line between them is 10 per cent of equity - at or above it FDI, below it FPI.

- Foreign exchange reserves have four components - foreign currency assets, gold, SDRs and the reserve tranche position with the IMF.
- Foreign currency assets are by far the largest component.
- India's reserves crossed US$ 700 billion for the first time in September 2024.
- India is among the top four reserve holders in the world.
- The reserve tranche position with the IMF is part of reserves - a favourite statement to plant as 'false' when it is true.

India's exchange rate regime – the sequence

Pre-1991: pegged, administered rate → 1991: two-step devaluation, 1 and 3 July → 1992: LERMS, a dual exchange rate → March 1993: rates unified into a managed float → August 1994: current account convertibility accepted under IMF Article VIII → 1997 and 2006: Tarapore Committees on capital account convertibility – not adopted fully

TRAP

India has current account convertibility, accepted under Article VIII in August 1994. India does NOT have full capital account convertibility – the two Tarapore Committees recommended a phased path and it was never completed. Any option claiming full convertibility of the rupee is wrong.

Trade policy, the WTO and India's trade agreements

Three separate things live here and you should keep them in separate boxes. One, India's own trade numbers and partners. Two, the multilateral body – the WTO. Three, the bilateral deals India signs. Each has its own set of one-liners.

- India's total exports in FY 2024-25 were a record US$ 820.93 billion - merchandise US$ 437.42 billion plus services US$ 383.51 billion.
- Merchandise imports in FY 2024-25 were US$ 720.24 billion.
- The United States has been India's largest trading partner in recent years; China is the largest source of India's imports.
- Crude petroleum is the single largest item in India's import basket.
- The WTO came into existence on 1 January 1995 under the Marrakesh Agreement of 1994, succeeding GATT 1947; headquarters at Geneva, decisions by consensus.
- The WTO had 166 members after Comoros and Timor-Leste joined at MC13 in Abu Dhabi in 2024.
- Foreign Trade Policy 2023 took effect on 1 April 2023 and is open-ended - it has no fixed end date - and is administered by the DGFT under the Ministry of Commerce and Industry.

India's recent trade agreements - name, partner, year

Agreement	Partner	Status
CEPA - Comprehensive Economic Partnership Agreement	United Arab Emirates	2022
ECTA - Economic Cooperation and Trade Agreement	Australia	2022
TEPA - Trade and Economic Partnership Agreement	EFTA bloc	Signed 2024, in force 2025; carries a US$ 100 billion investment commitment
CETA - Comprehensive Economic and Trade Agreement	United Kingdom	Signed July 2025
RCEP	ASEAN plus five	India walked out of negotiations in November 2019

CURRENT LINK

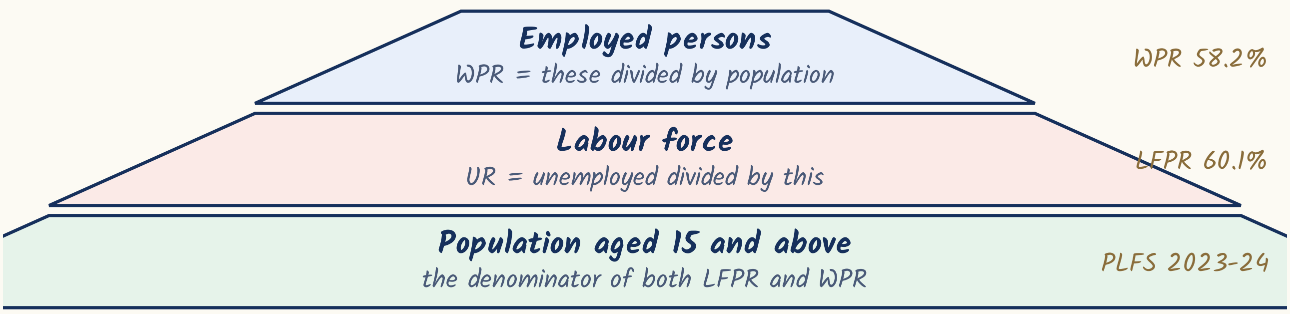
Current link, as of September 2026: India and the European Union announced the conclusion of their long-negotiated free trade agreement in late January 2026, widely reported as India's largest trade deal. Track its formal signature and ratification. On reserves, RBI weekly data put India's foreign exchange reserves at a record US$ 725.727 billion for the week ended 13 February 2026 - foreign currency assets US$ 573.603 billion, gold US$ 128.466 billion, SDRs US$ 18.924 billion. Reserve figures move every week, so carry the latest one you see and remember the four components rather than the number.

Employment and skills - definitions first, numbers second

Employment questions in RAS are definition questions dressed up as data questions. Before any number, be sure of three ratios. Labour Force Participation Rate is the labour force - workers plus those seeking work - as a share of population. Worker Population Ratio, also called the Work Participation Rate, is employed persons as a share of total population. Unemployment Rate is the unemployed as a share of the labour force, not of the population. Get those three denominators right and you cannot be tricked.

- Work Participation Ratio = employed persons divided by total population, expressed as a percentage.
- LFPR counts workers PLUS those seeking or available for work, as a share of population.
- Unemployment Rate = unemployed as a share of the LABOUR FORCE. Different denominator.
- About 90 per cent of India's workforce is in the unorganised or informal sector.
- The e-Shram portal, launched on 26 August 2021 by the Ministry of Labour and Employment, is the national database of unorganised workers, with over 30 crore registrations.
- e-Shram registers unorganised workers only. An option saying it covers organised-sector workers is wrong.

How LFPR, WPR and the unemployment rate nest



ASKED IN RAS

Asked in RAS Pre 2021 and again in RAS Pre 2024 - the Work Participation Ratio is defined as the ratio of employed persons to the total population. The wrong options offer the labour force to population ratio (that is LFPR) and unemployed to labour force (that is the unemployment rate). This is a pure definition mark, twice given away.

PLFS - where India's employment numbers come from

- The Periodic Labour Force Survey was launched by the NSO in April 2017, replacing the old quinquennial NSSO employment surveys.
- It gives annual estimates on usual status and shorter-period bulletins on current weekly status.
- PLFS 2023-24, usual status, persons aged 15 years and above: LFPR 60.1 per cent, Worker Population Ratio 58.2 per cent, unemployment rate 3.2 per cent.
- Female LFPR rose from 37.0 to 41.7 per cent in that round - the headline change.
- PLFS has since moved to calendar-year reporting.
- For calendar 2024 (PIB, usual status, 15 years and above): LFPR 59.6, WPR 57.7, unemployment 3.2 per cent; unemployment on current weekly status 4.9 per cent.
- If a question names a year, answer from that year's set. Do not mix the 2023-24 figures with the calendar-2024 figures.

India's skill development architecture - the match-the-following set

Body or scheme	Year	Key fact
Ministry of Skill Development and Entrepreneurship	9 November 2014	Carved out to coordinate skilling spread across more than twenty ministries
NSDC - National Skill Development Corporation	2008	A public-private partnership - Government 49 per cent, private sector 51 per cent
NSQF - National Skills Qualifications Framework	Notified December 2013	Ten levels, competency-based
National Skill Development Mission, PMKVY and the National Policy on Skill Development and Entrepreneurship	All on 15 July 2015	15 July is World Youth Skills Day - one date, three launches
NAPS - National Apprenticeship Promotion Scheme	2016	Promotes apprenticeship under the Apprentices Act, 1961

India's skill development architecture

Skill India – MSDE, 9 November 2014

NSDC, 2008
a PPP company
- Government 49%
- Private sector 51%
- Not govt-owned

NSQF, Dec 2013
qualification levels
- Ten levels
- Competency-based
- Credit transfer

15 July 2015
Youth Skills Day
- Skill Dev Mission
- PMKVY
- Skill Policy 2015

NAPS, 2016
apprenticeships
- Apprentices Act
- Under MSDE

ASKED IN RAS

Asked in RAS Pre 2018 – the Ministry of Skill Development and Entrepreneurship was created in 2014. Asked again in RAS Pre 2021 as statements on the skilling architecture, where the false statement was that NSDC is fully government-owned. It is not – the Government holds 49 per cent and the private sector 51 per cent. Learn 49:51 in that order.

YAAD RAKHO

Remember one date, three launches: 15 July 2015 – National Skill Development Mission, PMKVY and the National Policy on Skill Development and Entrepreneurship, all on World Youth Skills Day. And for the gig economy remember one law and one number: the Code on Social Security, 2020 first defined 'gig worker' and 'platform worker', and NITI Aayog's 2022 report projected the gig workforce rising from about 77 lakh in 2020-21 to 2.35 crore by 2029-30.

CURRENT LINK

Current link, as of September 2026: PLFS has shifted from the July-June survey year to calendar-year reporting, so two sets of headline ratios are now in circulation. PLFS 2023-24 (usual status, 15 years and above): LFPR 60.1, WPR 58.2, unemployment 3.2 per cent. Calendar 2024: LFPR 59.6, WPR 57.7, unemployment 3.2 per cent on usual status and 4.9 per cent on current weekly status. RPSC quotes whichever set is newest at the time of setting, so read the year in the question before you answer.

REVISION RECAP

1. Services give the largest share of gross value added, roughly 55 per cent; agriculture still employs the most people.
2. Total exports FY 2024-25 US$ 820.93 bn (merchandise 437.42 + services 383.51), merchandise imports 720.24 bn; USA largest trading partner, China largest import source, crude petroleum the biggest single import.
3. Electricity Act 2003 unbundled the sector, UDAY November 2015 for DISCOM finances, RDSS 2021; Panchamrit at COP26 - 500 GW non-fossil by 2030 and net zero by 2070; PM Surya Ghar 13 February 2024, 300 free units.
4. Golden Quadrilateral 5,846 km; North-South Srinagar-Kanyakumari, East-West Silchar-Porbandar, crossing at Jhansi; Bharatmala Phase-I 2017, about 34,800 km, through NHAI.
5. Eastern DFC Ludhiana-Sonnagar, Western DFC JNPT-Dadri, agency DFCCIL incorporated 2006; Vande Bharat 15 February 2019, Delhi-Varanasi; Kavach is the anti-collision system.
6. Sagarmala 2015 under Ports, Shipping and Waterways; Major Port Authorities Act 2021 replaced the 1963 Act; UDAN 21 October 2016, first flight Shimla-Delhi April 2017, capped fares plus viability gap funding.
7. Telecommunications Act 2023 replaced the Telegraph Act 1885 and the Wireless Telegraphy Act 1933; TRAI Act 1997; 5G launched 1 October 2022; PM Gati Shakti 13 October 2021, GIS platform with BiSAG-N, 16 ministries, nodal DPIIT, seven engines excluding power transmission; National Logistics Policy 17 September 2022.
8. NIP - Atanu Chakraborty task force, about Rs 111 lakh crore for FY2020-25, shown on the India Investment Grid.
9. HAM = 40 per cent government funding during construction; BOT (Toll) puts traffic risk on the concessionaire, BOT (Annuity) on the authority; TOT monetises built roads;

EPC is not a PPP model; VGF is with the Department of Economic Affairs and the Kelkar Committee 2015 reviewed PPPs.

10. Current account = goods plus invisibles; capital account = FDI, FPI, ECBs, banking capital. ECBs are capital account. The BoP always balances.
11. FDI means a lasting interest, conventionally 10 per cent or more of equity; below that it is FPI, the volatile hot money.
12. Reserves = foreign currency assets + gold + SDRs + reserve tranche position; crossed US$ 700 billion in September 2024; India is among the top four holders.
13. LERMS 1992, unified March 1993, current account convertibility under Article VIII in August 1994; Tarapore Committees 1997 and 2006 - full capital account convertibility not adopted.
14. WTO from 1 January 1995, Marrakesh Agreement 1994, Geneva, consensus decisions, 166 members after MC13 in 2024; FTP 2023 from 1 April 2023, open-ended, run by DGFT.
15. UAE CEPA 2022, Australia ECTA 2022, EFTA TEPA signed 2024 in force 2025, UK CETA signed July 2025; India exited RCEP talks in November 2019.
16. WPR = employed / population; LFPR = labour force / population; unemployment rate = unemployed / labour force.
17. PLFS from April 2017; 2023-24 usual status 15+: LFPR 60.1, WPR 58.2, unemployment 3.2; female LFPR up to 41.7 per cent.
18. About 90 per cent of workers are unorganised; e-Shram 26 August 2021, Ministry of Labour and Employment, over 30 crore registrations; the Code on Social Security 2020 first defined gig and platform workers.
19. MSDE created 9 November 2014; NSDC 2008 with 49:51 government-private equity; NSQF December 2013 with 10 levels; NSDM, PMKVY and the skill policy all on 15 July 2015; NAPS 2016.
20. NITI Aayog 2022 projected the gig workforce rising from about 77 lakh in 2020-21 to 2.35 crore by 2029-30.